

RETAIL BUSINESS PERFORMANCE & PROFITABILITY ANALYSIS

1. INTRODUCTION

Retail businesses generate large amounts of sales data. Analyzing this data helps a business understand which products and categories generate profit, which areas need improvement, and how sales change over time. This project analyzes the SuperStoreOrders dataset to study retail business performance and profitability.

2. ABSTRACT

The main objective of this project is to analyze retail transaction data and identify important patterns in sales and profit. The dataset contains 51,290 records from 2011 to 2014. The data was cleaned and analyzed using SQLite and Python. Tableau was then used to create an interactive dashboard for easier understanding of the results.

The analysis focuses on category and sub-category profitability, regional performance, customer segments, slow-moving products, and quarterly sales and profit trends.

3. TOOLS USED

SQLite / DB Browser for SQLite: Used for SQL queries, data checking and business analysis.

Python with Pandas: Used for data cleaning, calculations, grouping and trend analysis.

Tableau: Used to create the interactive dashboard and visualizations.

CSV: Used as the original and cleaned dataset format.

4. STEPS INVOLVED

1. Imported the SuperStoreOrders.csv dataset into SQLite.
2. Checked the dataset for missing values and duplicate records.
3. Cleaned Sales and Profit values using Python and converted them into numeric values.
4. Used SQL queries to analyze category, sub-category, region, customer segment and product performance.
5. Used Pandas to calculate sales, profit, profit margin, yearly and quarterly trends.
6. Created a Tableau dashboard with KPIs, charts and filters.

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5. KEY FINDINGS

Overall Performance: Total sales were 12,642,905 and total profit was approximately 1,469,034.82. The overall profit margin was 11.62%.

Category Performance: Technology recorded the highest sales of 4,744,691 and the highest profit of approximately 663,778.73. Office Supplies generated approximately 518,473.83 profit, while Furniture generated approximately 286,782.25 profit.

Sub-category Performance: Tables recorded a loss of approximately 64,083.39, making it an important sub-category for further review. Copiers and Phones were among the major profit contributors.

Quarterly Performance: Q4 had the highest sales at 3,881,878 and the highest profit at approximately 454,988.71.

Yearly Performance: Sales increased from 2,259,511 in 2011 to 4,300,041 in 2014. Profit increased from approximately 248,940.81 in 2011 to 504,165.97 in 2014.

Slow-moving Products: Products with very low quantities sold were identified for further review. However, low sales quantity alone does not prove that a product is overstocked.

6. TABLEAU DASHBOARD

An interactive Tableau dashboard was created to present the analysis. It contains Total Sales, Total Profit, Total Quantity and Overall Profit Margin KPIs. It also includes Sales by Category, Profit by Category, Region Performance, Sales by Quarter and Profit by Quarter. Filters were provided for Region, Category and Quarter.

7. CONCLUSION

The project demonstrates how retail transaction data can be transformed into useful business information using SQL, Python and Tableau. The analysis helps identify profitable categories, loss-making sub-categories, regional patterns and quarterly trends. These findings can support future decisions related to product performance, pricing and sales planning.

The dataset does not contain inventory-level or inventory-days information. Therefore, inventory turnover and overstocking should be analyzed separately when such data is available.